

【物理化学特論】 3. Square well potential

No. : _____ Name : _____

Answer the following the questions to solve the Schrödinger equation for a particle in a one-dimensional box.

- (1) Write the Hamiltonian for a particle in a box, where the potential is given as

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & otherwise \end{cases}$$

- (2) Determine the solution (wave function $\psi(x)$) in $x < 0$ and $x > L$.

- (3) Determine the general solution in $0 < x < L$, where the energy of the particle is assumed to be positive.

- (4) Deduce the eigen energies E_n ($n=1, 2, 3, \dots$) with the boundary condition at $x = 0$ and L .

- (5) Deduce the wave functions $\psi_n(x)$ with the normalization conditions.

- (6) Calculate the lowest excitation energy for an electron ($m_e=9.1 \times 10^{-31}$ kg) confined in a slit with 0.1 nm width (in eV unit).

- (7) Calculate the wavelength for the excitation energy of (6).